B.Sc. (Hons) Part-IV Examination 2022

PHYSICS

Course: PH401

Quantum Mechanics

Full Marks: 80 **Time: 4 hours**

Answer ANY FIVE questions

1. (a) Discuss how matrices represent operators and wave functions in the matrix formulation of quantum mechanics. 6

   (b) Explain, with an example, the role played by unitary operators in transformation theory. Why is it important that the operators are unitary? 8+2

2. (a) Discuss parity symmetry and parity operator. Show that parity is conserved in case of a spherically symmetric potential. 5

   (b) Prove that $\vec{J} \times \vec{J} = i\hbar\vec{J}$ for the angular momentum operator $\vec{J}$. 3

   (c) Obtain the Pauli spin matrices. 8

3. (a) Define angular momentum ladder operators. Why are they called so? 3

   (b) Prove that $L_+Y_l^m$ *and* $L_-Y_l^m$ are simultaneous eigenfunctions of L^2 *and* L_z 8

   (c) Define Clebsch-Gordon coefficients and discuss its necessity. 5

4. (a) For elastic scattering, deduce expressions for scattering amplitude and differential cross section in the first Born approximation. 6

   (b) Obtain the validity criteria of the first Born Approximation. 3

   (c) Calculate the differential cross section in the first Born Approximation for a Coulomb potential $V(r) = \frac{Z_1 Z_2 e^2}{r}$, where $Z_1 e$ and $Z_2 e$ are the charges of the projectile and target particles, respectively. 4

   (d) What is phase shift δ_l? What is its physical significance? 1+2

5. (a) For an infinite one-dimensional square well of width a, the unperturbed wavefunctions are $\psi_n^0 = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)$ and energies are $E_n^0 = \frac{\pi^2\hbar^2 n^2}{2ma^2}$. Find the first order correction to the energies if the floor of the well is raised to V_0 for $0 < x < \frac{a}{2}$. 4

   (b) In a three-dimensional infinite cubical well (particle in a cubical box) the energy eigenvalues are $E_{n_x,n_y,n_z} = \frac{\pi^2\hbar^2}{2ma^2}\left(n_x^2 + n_y^2 + n_z^2\right)$ and the first excited states ψ_{211}, ψ_{121} and ψ_{112} are degenerate. Show how this degenarcy is lifted when the following perturbing potential is applied: 9

   $$H' = \begin{cases} V_0 & 0 < x < \frac{a}{2}, 0 < y < \frac{a}{2} \\ 0 & \text{elsewhere} \end{cases}$$

   (c) Usually when a perturbation is applied, degeneracy is lifted. Why? Can you think of a case where, even after applying a perturbation, the degeneracy will survive. 2+1

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6. (a) Use a triangular trial wavefunction given by 7+1

$$\psi(x) = \begin{cases} Ax & 0 \le x \le a/2 \\ A(a-x) & a/2 \le x \le a \\ 0 & \text{elsewhere} \end{cases}$$

to estimate the ground state energy of a one-dimensional infinite square well. Compare your result with the true ground state energy $E_{gs} = \pi^2\hbar^2/2ma^2$.

(b) In the WKB approximation, starting from the asymptotic solutions for a right-hand upward-sloping turning point at $x = 0$: 8

$$\psi(x) \cong \begin{cases} \frac{1}{\sqrt{p(x)}}\left[Be^{\frac{i}{\hbar}\int_x^0 p(x')dx'} + Ce^{-\frac{i}{\hbar}\int_x^0 p(x')dx'}\right], & \text{if } x < 0 \\ \frac{1}{\sqrt{|p(x)|}}\left[De^{-\frac{i}{\hbar}\int_0^x |p(x')|dx'}\right], & \text{if } x > 0 \end{cases}$$

Obtain the connection formulas that relate B, C and D.

7. (a) Consider a two-level system with states a and b, such that $\Psi(t=0) = c_a\psi_a + c_b\psi_b$. Obtain the expressions: $\dot{c}_a = -\frac{i}{\hbar}H'_{ab}e^{-i\omega_0 t}c_b$ and $\dot{c}_b = -\frac{i}{\hbar}H'_{ba}e^{-i\omega_0 t}c_a$, where $\omega_0 = (E_b - E_a)/\hbar$. 8

(b) We have learnt that the calculation of spontaneous emission rates reduces to the evaluation of matrix elements of the form $\langle\psi_b|\vec{r}|\psi_a\rangle$. Obtain the selection rules for Δm and Δl. 8

8. (a) What was the motivation for the Dirac equation? 4

(b) Explain how Dirac's equation leads to the concept of intrinsic spin of electron. 6

(c) Discuss the physical significance of the negative energy solution of the Dirac equation. 6

$P(x) = \alpha^{3/2}\hbar\sqrt{-x}$

$\int_x^0 P(x')dx' = \frac{2}{3}\hbar(-\alpha x)^{3/2}$

$H\psi = E\psi$

$\hbar\omega = \pm\sqrt{}$

$-\frac{\omega^2}{c^2} + k^2 + \frac{m^2c^2}{\hbar^2}$

$\psi(\alpha) = Ae^{-b\alpha^2}$

$\psi(\alpha) = \langle\psi|\psi\rangle = 1$

$\frac{1}{2}m\omega^2x^2$

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B.Sc. (Hons.) Part-IV Examination-2021

Physics

Course No.: PH401

Course Title: Quantum Mechanics

Full Marks: 80 **Duration:** 04 hours

(N.B. *Figures in the right margin indicate marks. Answer any FIVE questions.*)

1. a) What do you mean by state vectors? Explain. 3
   b) What is a linear operator? Mention its characteristics. 3
   c) Define matrix elements of an operator. What are the significance of the matrix elements? 3
   d) Write a short note on Hilbert space. 4
   e) Show that the off-diagonal matrix elements of an operator are oscillatory in the energy representation. What does it mean? 3

2. a) Discuss how the evolution of a physical system is visualized in the Schrödinger, Heisenberg and Interaction pictures. 9
   b) Prove that $\vec{L} \times \vec{L} = i\hbar\vec{L}$ for the angular momentum operator $\vec{L}$. 3
   c) Find the rotational energy levels of a diatomic molecule. Draw a diagram, showing first four energy levels. 3+1

3. a) If $|jm\rangle$ is a simultaneous eigenstate of J^2 and J_z, obtain the allowed values of j and m. 8
   b) Obtain the matrices for J_x, J_y, J_z and J^2 operators for $j = 1$. 8

4. a) Explain briefly the effects of attractive and repulsive potentials on phase shifts. 3
   b) Find an expression for total inelastic scattering cross-section. 4
   c) Consider the elastic scattering of 50 MeV neutrons from a nucleus. The phase shafts measured in this experiment are $\delta_0 = 98°, \delta_1 = 82°, \delta_2 = 63°, \delta_3 = 40°, \delta_4 = 15°, \delta_5 = 3°$; all other phase shifts are negligible (i.e. $\delta_l \simeq 0$ for $l \geq 6$). Find the total cross-section. 3
   d) Consider the scattering of a particle of mass m from a hard sphere potential: $V(r) = \infty$ for $r < a$ and $V(r) = 0$ for $r > a$. 4+2
      (i) Calculate the total cross-section in the high and low energy limits.
      (ii) Explain why the classical value of cross-section is not obtained in question (i).

5. a) Obtain expression for the energy shifts and wavefunctions for non-degenerate states in the first order stationary perturbation theory. 8
   b) For an infinite one-dimensional square well of width a, the unperturbed wave functions are $\psi_n^0 = \sqrt{\frac{2}{a}} sin\left(\frac{n\pi}{a}x\right)$ and energies are $E_n^0 = \frac{\pi^2\hbar^2n^2}{2ma^2}$. Fined the first order corrections to the energies for a delta-function bump in the middle of the box: $H' = \beta\delta(x - a/2)$. 6
   c) Can you guess why the correction is zero for even n? 2

6. a) Calculate the ground state energy of a one-dimensional harmonic oscillator in the variational method using a suitable trial function. 6
   b) What are the classical turning points in the WKB approximation and why are they called so? 4
   c) Obtain, with necessary assumptions, the WKB asymptotic solutions. 6

7. a) Show that for a sinusoidal perturbation $H'(\vec{r}, t) = V_{ab}cos(\omega t)$, the transition probability of a transition from a to b in given by : $P_{a\to b}(t) \cong \frac{|V_{ab}|^2}{\hbar^2}\frac{sin^2[(\omega_0-\omega)t/2]}{(\omega_0-\omega)^2}$. 8
   b) Plot this against t and ω. What conclusions can you make about transitions from these two plots 4+4

8. a) Obtain the plane wave solutions of the Dirac's relativistic equation. 10
   b) Explain how Dirac's equation leads to the concept of intrinsic spin of electron. 4
   c) Prove that the operator $C\vec{\alpha}$ corresponds to particle velocity. 2

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B.Sc. (Hons.) Part-IV Examination 2020
(According to old syllabus)
PHYSICS
Course: **PH401**
(Quantum Mechanics)

ull Marks: **100** Duration: **4 hours**

*The figures in the right margin indicate full marks. Answer Any **FIVE** questions*

(a) Define matrix element of an operator. What is its significance? 4
(b) Explain, with an example, the role played by unitary operators in transformation theory. 10+2
(c) Discuss the essential features of energy representation. 4

(a) Obtain the energy eigenvalues of a one-dimensional harmonic oscillator. How does it differ from the classical result? 10+3
(b) Find the matrices for $\hat{x}$ and $\hat{p}$ of a one-dimensional harmonic oscillator. 7

(a) Obtain Pauli spin matrices. 10
(b) Derive the matrices for $\hat{J}_x$, $\hat{J}_y$, $\hat{J}_z$ and $\hat{J}^2$ when $j = 2$. 10

(a) Introduce the scattering amplitude. Derive how is it related to the cross-section. 8
(b) Write down an expression for the scattering amplitude in the Born approximation. Obtain the differential cross-section in the Born Approximation for a Yukawa potential $V(r) = V_0 e^{-ar}/r$. Show that, at an appropriate limit, this tends to the Rutherford formula. 10+2

(a) What are the conditions on a Hamiltonian for perturbation theory to be valid? 3
(b) For an particle in a cubical box of sides a, the unperturbed wavefunctions are 3+11
$\psi^0_{n_x,n_y,n_z}(x,y,z) = \left(\frac{2}{a}\right)^{3/2} sin\left(\frac{n_x\pi}{a}x\right) sin\left(\frac{n_y\pi}{a}y\right) sin\left(\frac{n_z\pi}{a}z\right)$ and energies are $E^0_{n_x,n_y,n_z} = \frac{\pi^2\hbar^2}{2ma^2}(n_x^2 + n_y^2 + n_z^2)$. Explain how the first excited states are degenerate. When a perturbing potential of the form $H' = \begin{cases} V_0 & 0 < x < a/2,\ 0 < y < a/2 \\ 0 & elsewhere \end{cases}$ is applied, show how this degeneracy is broken.
(c) How is the breaking of degeneracy connected to symmetry? 3

(a) State and discuss variational principle. What criteria should be used to select a trial wave function? 10
(b) Calculate the ground state energy of a hydrogen atom using a trial wave function of the form $\psi = Ne^{-\alpha r}$. 8
(c) What is resonance scattering? Explain. 2

(a) For the solutions far away from the right-hand turning point- 12

$$\Psi(x) = \begin{cases} \frac{1}{\sqrt{p(x)}}\left[B\ exp\left\{\frac{i}{\hbar}\int_x^0 p(x')dx'\right\} + C\ exp\left\{-\frac{i}{\hbar}\int_x^0 p(x')dx'\right\}\right] & for\ 0 < x \\ \frac{1}{\sqrt{|p(x)|}} D\ \exp\left[-\frac{i}{\hbar}\int_0^x |p(x')|dx'\right] & for\ x < 0 \end{cases}$$

obtain the connection formula between B, C and D.
(b) Obtain the energy spectrum for a half-harmonic oscillator 8

$$V(x) = \begin{cases} \frac{1}{2}m\omega^2x^2 & x > 0 \\ \infty & elsewhere. \end{cases}$$

(a) Solve Klein-Gordon equation for a free particle and obtain the energy eigenvalue. 7
(b) Obtain the matrices for $\bar{\alpha}$ and β from Dirac's relativistic wave equation. 8
(c) Write a note on Dirac's hole theory. 5

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B.Sc. (Hons.) Part-IV Examination 2019
PHYSICS
Course: PH401
(Quantum Mechanics)

Full Marks: **80** Duration: **4 hours**

The figures in the right margin indicate full marks. Answer any **FIVE** questions.

1.	(a)	What do you mean by state vectors? Explain.	3
	(b)	What is a linear operator? Mention its characteristics.	3
	(c)	Define matrix elements of an operator. What are the significances of the matrix elements?	3
	(d)	Write a short note on Hilbert space.	4
	(e)	Show that the off-diagonal matrix elements of an operator are oscillatory in the energy representation. What does it mean?	3
2.	(a)	Obtain the equation of motion for an operator in the interaction picture.	6
	(b)	Compare among Schrödinger, Heisenberg, and interaction representations.	5
	(c)	Discuss parity symmetry and parity operator. Show that parity is conserved in the case of a spherically symmetric potential.	5
3	(a)	Solve the energy eigenvalue problem of a one-dimensional harmonic oscillator and obtain the matrices for creation, annihilation, and number operators.	10
	(b)	Consider a spin $\frac{1}{2}$ system and derive the Pauli spin matrices.	6
4	(a)	What is partial wave? Explain.	3
	(b)	Obtain an expression for the scattering amplitude in the partial wave expansion.	7
	(c)	What is the significance of the (partial) phase shifts, δ_l?	3
	(d)	When is the partial wave expansion useful?	3
5	(a)	Obtain the first order corrections to the energy and wavefunction in the non-degenerate perturbation theory.	6
	(b)	Obtain the first order correction to the energy of a particle in a one-dimensional infinite square well of width a when half of its floor (from 0 to $a/2$) is raised by V_0.	2
	(c)	What criteria should be used when we choose a trial wavefunction?	1
	(d)	Use the variational method to obtain an estimate of the ground state energy of a one-dimensional harmonic oscillator.	7
6.	(a)	What do you understand by the term 'classical turning points' in the context of the WKB approximation?	2
	(b)	In the WKB method, obtain the solutions far from the turning points.	7
	(c)	Why are these solutions not valid at the turning points?	2
	(d)	Write down briefly the steps required to connect the solutions across a turning point in the WKB approximation.	5
7.	(a)	In time-dependent perturbation theory, consider a two-level system labeled a and b, obtain the expressions $\dot{c}_a = -\frac{i}{\hbar} H'_{ab} e^{-i\omega_0 t} c_b$ and $\dot{c}_b = -\frac{i}{\hbar} H'_{ab} e^{-i\omega_0 t} c_a$.	6
	(b)	From the above relations, obtain first order expressions for $c_a(t)$ and $c_b(t)$. If $c_a(0) = 1$ and $c_b(0) = 0$, for a sinusoidal perturbation, obtain an expression for a transition from a to b at time t.	6
	(c)	Plot the expression for $P_{a\to b}(t)$, obtained in question (b), against ω and t and discuss the consequences.	4
8.	(a)	Solve free-particle Dirac relativistic wave equation.	10
	(b)	Set-up Klein-Gordon equation and obtain the energy eigenvalue. Comment on the result.	6

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B.Sc. (Honours) Part-IV Examination 2018

PHYSICS

Course: **PH401**

Quantum Mechanics

[Answer any FIVE questions]

Duration: 4 hours | Full Marks: 100

1. a) Discuss how matrices represent operators and wave functions in the matrix formulation of quantum mechanics. 8
   b) Explain, with an example, the role played by unitary operators in transformation theory. Why is it important that the operators are unitary? 10+2

2. a) Obtain the energy spectrum of a one-dimensional harmonic oscillator. How is this different from the classical case? 8+2
   b) What are Clebsch-Gordon coefficients? Explain with an example. 5
   c) Show that $[J_x, J_y] = i\hbar\varepsilon_{xyz} J_z$. 5

3. a) Consider the eigenket $|jm\rangle$ and show that the allowed values of m are $m = -j,\ -j+1, \ldots, +j$. 5
   b) Prove that the non-vanishing elements of the raising and lowering angular momentum operators $J_\pm$ are given by 15

$$\langle j, m+1|J_\pm|j,m\rangle = \sqrt{(j \mp m)(j \pm m + 1)}\hbar$$

4. a) In a partial wave expansion, obtain an expression for the scattering amplitude in terms of the partial phase shifts. 7
   b) When is the partial wave analysis useful? 3
   c) Obtain the differential scattering cross-section for the potential $V(r) = V_0 \frac{e^{-ar}}{r}$ in the Born Limit. 6
   d) Explain the term 'scattering' within the context of quantum mechanics. What is the significance of total scattering cross-section? 4

5. a) Describe the utility of perturbation treatment. 3
   b) Obtain an expression for the energy shifts and wave functions for non-degenerate states in the first-order perturbation theory. 11
   c) What do you mean by 'Lifting of degeneracy' in the context of stationary degenerate perturbation theory? Illustrate with an example. 6

6. a) Show that for a sinusoidal perturbation of the form, $H'(\bar{r}, t) = \vec{V}_{ab} Cos(\omega t)$, the transition probability from states 'a' to 'b' is given by $P_{a\to b}(t) \cong \frac{|V_{ab}|^2}{\hbar^2} \frac{Sin^2\left[\frac{(\omega_0-\omega)t}{2}\right]}{(\omega_0-\omega)^2}$. 10
   b) Plot this probability function against 't' and 'w'. What conclusions can you make about the transitions from these two plots? 5+5

7. a) Show that for any normalized wave function ψ, $E_{gs} \leq \langle\psi|H|\psi\rangle$. How can we use this to obtain the ground state wave functions in the variational method? 7+3
   b) What is the 'classically allowed' region in the WKB approximation? Show that, in this region, the wave function can be expressed as $\psi(x) \cong \frac{C}{\sqrt{P(x)}} e^{\pm\frac{i}{\hbar}\int P(x)dx}$. 10

8. a) What was the motivation for the Dirac equation? 4
   b) Derive the Dirac equation and show how 4×4 matrices are needed. How are these matrices related? 12
   c) Discuss the physical significance of the negative energy solution of the Dirac equation. 4

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B.Sc. (Honours) Part-IV Examination 2017

Physics

Course: PH401

Quantum Mechanics

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Full Marks: 100 **Time: 4 hours**

[Answer ANY FIVE questions]

1.	a)	Describe how state vectors and operators are represented in Hilbert space.	8
	b)	Discuss how the evolution of a physical system is visualized in the Schrodinger, Heisenberg and Interaction (Dirac) pictures.	12
2.	a)	Solve the energy eigen-value problem of a one-dimensional harmonic oscillator.	8
	b)	Obtain the matrix for the Hamiltonian of the linear harmonic oscillator.	4
	c)	What is the origin of the zero-point energy?	2
	d)	Set up and solve the quantum mechanical equation of motion for momentum operator of a free particle. Comment on the solution.	6
3.	a)	Mention the necessity of Clebsch-Gordon coefficients. What are the selection rules? Derive the selection rules.	3+1+4
	b)	Obtain the matrices for J_x, J_y, J_z, and J^2 operators for j=2.	10
	c)	Define angular momentum ladder operators. Why are they called so?	2
4.	a)	For a scattering from a rigid sphere, calculate the cross-section in the high and low energy limits.	8
	b)	For a square well potential, obtain an expression for the phase shift in the low energy limit.	8
	c)	In this context, explain the phenomenon of resonance scattering.	4
5.	a)	What are the conditions on the Hamiltonian for perturbation theory to be applicable?	3
	b)	In degenerate perturbation theory, consider a system with a two-fold degeneracy and obtain an expression for the first order corrections to the energies. What do we need to know to calculate these?	8+2
	c)	For an infinite one-dimensional square well of width a, the unperturbed wave-functions are $\psi_n^0 = \sqrt{\frac{2}{a}} sin\left(\frac{n\pi}{a}x\right)$ and energies are $E_n^0 = \frac{\pi^2\hbar^2n^2}{2ma^2}$. Find the first order corrections to the energies for a delta-function bump in the middle of the box: $H' = \beta\delta\left(x - \frac{a}{2}\right)$.	5
	d)	Can you guess why the correction is zero for even n?	2
6.	a)	State and prove the variational principle.	7
	b)	What criteria should be used when we choose a trial wave-function?	3
	c)	Use the variational method to obtain an estimate of the ground state energy of a one dimensional harmonic oscillator.	10
7.	a)	Obtain the validity criterion of the WKB approximation.	4
	b)	Derive the WKB asymptotic solutions. What happens at the turning point?	6+2
	c)	Calculate the energy levels of a simple harmonic oscillator using the WKB-derived quantization rule.	8
8.	a)	Solve Klein-Gordon equation for a free particle.	6
	b)	Obtain the matrices for $\bar{\alpha}$ and β from Dirac's relativistic wave equation.	8
	c)	Show that the operator $c\bar{\alpha}$ corresponds to particle velocity.	4
	d)	Discuss the significance of negative energy particles in Dirac's hole theory.	2

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B.Sc. (Honours) Part-IV Examination 2016

Physics

Course: PH401

Quantum Mechanics

Full Marks: 100 Time: 4 hours

[*Answer ANY FIVE*]

1. (a) Define parity. What does parity conservation mean? Show that the eigenvalues of the parity operator are ± 1. 6
   (b) Write a short note on density matrix and density operator. 5
   (c) Obtain the equation of motion for an operator in the interaction picture. Compare the different representations. 6+3

2. (a) What is the utility of unitary transformations in quantum mechanics? Illustrate with an example. 5
   (b) Deduce an expression for zero-point energy of a one-dimensional harmonic oscillator. What is its significance? 7
   (c) Show that the expectation values of position and momentum are zero for a one-dimensional harmonic oscillator. 8

3. (a) Prove that $\vec{J} \times \vec{J} = i\hbar\vec{J}$ for the angular momentum operator $\vec{J}$. 5
   (b) Obtain the Pauli spin matrices. 8
   (c) Derive the matrices for J_x, J_y, J_z, and J^2 operators, when $j = \frac{3}{2}$. 7

4. (a) In the partial wave expansion, obtain an expression for the scattering amplitude in terms of the partial phase shifts. 7
   (b) When is the partial wave expansion useful? 3
   (c) Obtain the differential scattering cross-sections for scattering from a Yukawa potential $V(r) = V_0 \frac{e^{-ar}}{r}$ in the Born approximation. 6
   (d) Consider the limit where this tends to the Coulomb potential and show that this leads to the Rutherford scattering cross-section. 4

5. (a) Obtain an expression for the energy shifts and wavefunctions for non-degenerate states in the first order stationary perturbation theory. 10
   (b) A particle in a one-dimensional well of width a is perturbed by the potential 6

$$V = V_0 \text{ for } 0 < x < a/2$$
$$= 0 \text{ for } a/2 < x < a$$

   Find the first order correction to the energy eigen value.
   (c) What do you mean by "lifting of degeneracy" in the context of stationary degenerate perturbation theory? 4

6. (a) Show that for a sinusoidal perturbation $H'(\vec{r}, t) = V_{ab}\cos(\omega t)$, the transition probability from a to b is given by: 10

$$P_{a\to b}(t) \cong \frac{|V_{ab}|}{\hbar^2} \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2}$$

   (b) Plot this against t and ω. What conclusions can you make about the transitions from these two plots? 5+5

7. (a) Show that for any normalized wavefunction ψ, $E_{gs} \leq \langle\psi|H|\psi\rangle$. How can we use this to obtain ground state wavefunctions in the variational method? 7+3
   (b) What is the "classically allowed" region in the WKB approximation? Show that, in this region, the wavefunction can be written as $\psi(x) \cong \frac{C}{\sqrt{p(x)}} e^{\pm\frac{i}{\hbar}\int p(x)dx}$. 3+7

8. (a) Obtain the equation of continuity from the Klein-Gordon equation. Compare it with that obtained for the non-relativistic wave equation. 8
   (b) Solve the free particle Dirac equation. 12

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B.Sc. (Honours) Part-IV Examinations 2015

Physics

Course: **PH401**

Quantum Mechanics

Full marks: **100** Duration: **4** hours

N.B. Figures in the right margin indicate full marks. Answer any FIVE questions.

			marks
1.	a)	Define state vectors. Give an account of the Hilbert space.	2+4
	b)	Obtain the equation of motion for a dynamical variable in the Heisenberg representation. Compare it with the one in Schrödinger representation.	8
	c)	What is parity? Explain with example. Show that, for a central potential the parity operator commutes with the Hamiltonian.	3+3
2.	a)	Show that the energy eigenvalues of a one-dimensional harmonic oscillator is given by $E_n = (n + ½)\,h\upsilon$, where the symbols have their usual meanings.	7
	b)	Obtain the matrices for creation and destruction operators. What are the significances of these operators?	9
	c)	Write down the essential features of interaction and energy representations.	4
3.	a)	For a particular value of j, show that the number of possible eigenstates is given by (2j+1).	5
	b)	Derive the matrices for the operators J^2, J_x, J_y, J_z, when j = 2.	10
	c)	Define Clebsch-Gordan coefficients and discuss their necessity.	5
4.	a)	Write down the Born expansion. What is the Born approximation and under what condition is it valid?	2+4
	b)	Show that the sum effect of a real central potential on the *l*th partial wave is to introduce a phase shift of $2\delta_l$ in the outgoing spherical wave.	9
	c)	Why is it convenient to use the partial wave expansion in the description of scattering?	5
5.	a)	Formulate the first-order stationary perturbation theory for degenerate energy levels and explain how the degeneracy is lifted.	10
	b)	In a three-dimensional infinite cubical well (particle in a cubical box) the energy eigenvalues are $E_{n_x,n_y,n_z} = \frac{\pi^2\hbar^2}{2ma^2}\left(n_x^2 + n_y^2 + n_z^2\right)$ and the first excited states ψ_{211}, ψ_{121} and ψ_{112} are degenerate. Show how this degenarcy is lifted when the following perturbing potential is applied: $H' = \begin{cases} V_0 & 0 < x < \frac{a}{2}, 0 < y < \frac{a}{2} \\ 0 & elsewhere \end{cases}$.	10
6.	a)	Obtain an expression for the wave-function coefficients in first-order time dependent perturbation theory. [You may use a two-level system for simplicity.] How can this be used to describe transitions?	7+3
	b)	Find the transition probability for a harmonic perturbation. Show that only terms with $\omega = \omega_0$ will contribute significantly. [Here, as usual, $H_0\psi = E_0\psi = \hbar\omega_0\psi$.]	7+3
7.	a)	Calculate the ground state energy of a one-dimensional harmonic oscillator in the variational method using a suitable trial function.	6
	b)	What are the classical turning points in the WKB approximation and why are they called so? Obtain, with necessary assumptions, the WKB asymptotic solutions.	2+8
	c)	Outline the steps required to connect the two solutions across the turning points.	4
8.	a)	Obtain the plane wave solutions of the Dirac's relativistic wave equation.	10
	b)	Solve Klein-Gorclon equation and show that the energy is given by $\hbar\omega = \pm(\hbar^2c^2k^2 + m^2c^4)^{\frac{1}{2}}$, where the symbols have their usual meanings.	6
	c)	Explain how Dirac's equation leads to the concept of intrinsic spin of electron.	4

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